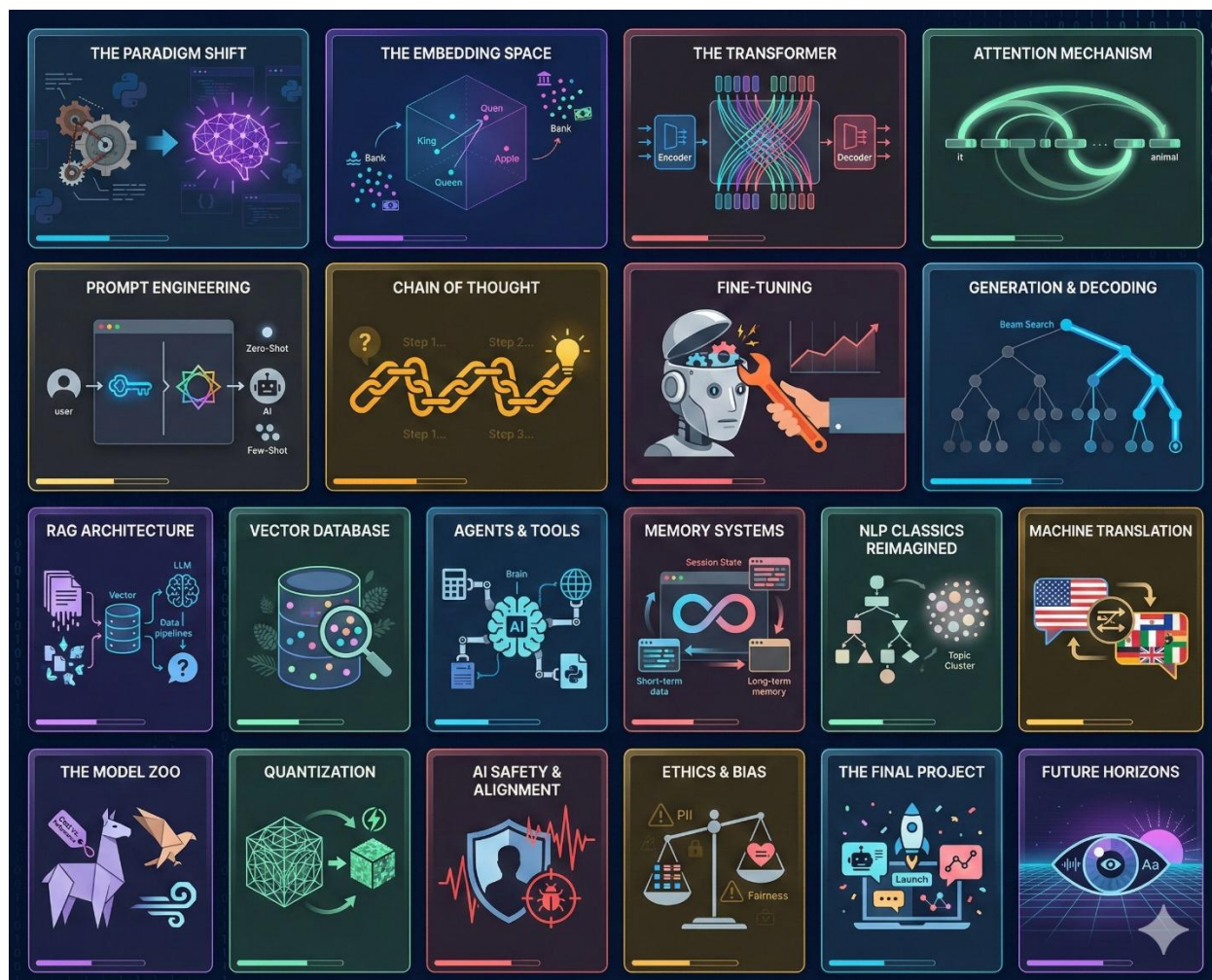




| Week | Dates          | Module               | Topic                                                                                        | Due Date                  |
|------|----------------|----------------------|----------------------------------------------------------------------------------------------|---------------------------|
| 1    | Jan 12 – 18    | <b>Mod 0 &amp; 1</b> | <b>Course Intro &amp; The LLM-First World</b><br>Pre-Transformer vs. Post-Transformer<br>NLP | Mon, Jan 19*<br>(MLK Day) |
| 2    | Jan 19 – 25    | <b>Mod 2</b>         | <b>Text Representation</b><br>From TF-IDF to Contextual<br>Embeddings                        | Mon, Jan 26               |
| 3    | Jan 26 – Feb 1 | <b>Mod 3</b>         | <b>The Engine: Transformers</b><br>Self-Attention, Encoders, & Decoders                      | Mon, Feb 2                |
| 4    | Feb 2 – 8      | <b>Mod 4</b>         | <b>Prompt Engineering</b><br>Zero-shot, CoT, & DSPy                                          | Mon, Feb 9                |

|           |                   |                       |                                                                              |                                                     |
|-----------|-------------------|-----------------------|------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>5</b>  | Feb 9 – 15        | <b>Mod 5</b>          | <b>Fine-Tuning (Specialization)</b><br>NER & Classification on the Edge      | Mon, Feb 16                                         |
| <b>6</b>  | Feb 16 – 22       | <b>Mod 6</b>          | <b>Generation &amp; Summarization</b><br>Decoding Strategies & LoRA/PEFT     | Mon, Feb 23                                         |
| <b>7</b>  | Feb 23 – Mar 1    | <b>Mod 7</b>          | <b>Retrieval Augmented Generation (RAG)</b><br>Vector DBs & Grounding        | Mon, Mar 2                                          |
| <b>8</b>  | Mar 2 – 8         | <b>Mod 8</b>          | <b>Chatbots &amp; Agents</b><br>Memory, Tools, and Agent Frameworks          | <b>Sun, Mar 8*</b><br>(Due before Break)            |
| <b>--</b> | <b>Mar 9 – 15</b> | <b>BREAK</b>          | <b>SPRING BREAK</b>                                                          | No Due Dates                                        |
| <b>9</b>  | Mar 16 – 22       | <b>Mod 9 &amp; 10</b> | <b>Syntax &amp; Topic Modeling</b><br>Knowledge Graphs & Semantic Clustering | Mon, Mar 23                                         |
| <b>10</b> | Mar 23 – 29       | <b>Mod 11</b>         | <b>Machine Translation</b><br>NMT & Multilingual Models                      | Mon, Mar 30                                         |
| <b>11</b> | Mar 30 – Apr 5    | <b>Mod 12</b>         | <b>The LLM Ecosystem &amp; MLOps</b><br>Quantization, Edge AI, & Cost        | <b>Tue, Apr 7*</b><br>(Adjusted for Spring Holiday) |
| <b>12</b> | Apr 6 – 12        | <b>Mod 14</b>         | <b>Ethics, Safety &amp; Alignment</b><br>Bias, Jailbreaking, & Red Teaming   | Mon, Apr 13                                         |
| <b>13</b> | Apr 13 – 19       | <b>Mod 15</b>         | <b>Multimodality &amp; Future Trends</b><br>Vision-Language Models (VLM)     | Mon, Apr 20                                         |
| <b>14</b> | Apr 20 – 26       | <b>Project</b>        | <b>Final Project Sprint 1</b><br>Implementation & Prototyping                | Mon, Apr 27                                         |
| <b>15</b> | Apr 27 – May 3    | <b>Project</b>        | <b>Final Project Sprint 2</b><br>Evaluation & Refinement                     | Mon, May 4                                          |
| <b>16</b> | May 4 – 10        | <b>Mod 16</b>         | <b>Course Wrap-Up</b><br>Final Presentations                                 | <b>Fri, May 8</b>                                   |